

PYTHON FOR BEGINNERS · DAY 6

Strings

Working with text like a pro

Instructor: Homaira Mohammadi

```
word = "Python"
print(word[0])
print(word.upper())
print(word[::-1])

# P PYTHON nohtyP
```

TODAY'S GOALS

By the end of today, you can…

- Get any letter with its **index**
- Use string **methods**
- Format output with **f-strings**
- Cut out parts with **slicing**
- Check text with **in**
- Build a palindrome checker 🔁

WHAT WE ALREADY KNOW

A string is text in quotes

```
name = "Ali"
print("Hi " + name)  # join
print("ha" * 3)      # repeat

for letter in name:
    print(letter)
```

+ joins strings

***** repeats a string

A **for** loop goes through each letter

Today: get **one** letter, or a **piece** of the text.

INDEXING

Every letter has a position number

P	y	t	h	o	n
0	1	2	3	4	5
-6	-5	-4	-3	-2	-1

```
word[0] → P
```

```
word[-1] → n
```

```
len(word) → 6
```

SLICING ✂️

text[start:stop] — cut out a piece

word = "Python"	Result	Meaning
word[0:3]	Pyt	index 0 up to (not incl.) 3
word[:4]	Pyth	from the start
word[2:]	thon	to the end
word[-3:]	hon	the last 3
word[::-1]	nohtyP	reversed!

Just like range(), the stop position is not included.

STRING METHODS 🛠️

Ready-made tools for text

s = "hello World"	Result	Job
s.upper()	HELLO WORLD	all capitals
s.lower()	hello world	all small letters
s.title()	Hello World	capital first letters
s.replace("World", "Ali")	hello Ali	swap text
s.count("o")	2	how many times
s.find("W")	6	where it starts

EXAMPLE

Tidy up what the user types

Code

```
raw = "  homaira mohammadi  "
name = raw.strip().title()
print("Hello,", name)
```

Output

```
Hello, Homaira Mohammadi
```

strip() removes spaces at the start and end

Methods can be **chained**: strip, then title

TWO MORE TOOLS

split() and in

split() — break apart

```
s = "Python is fun"
words = s.split()
print(words)
print(len(words))
```

```
['Python', 'is', 'fun']
3
```

in — is it inside?

```
email = "ali@mail.com"
if "@" in email:
    print("Valid ✅")
```

```
Valid ✅
```

⚠️ REMEMBER

Methods don’t change the original

They give back a **new** string. Store it to keep it.

```
text = "hello"
text.upper()
print(text)  # hello ❌

text = text.upper()
print(text)  # HELLO ✅
```

F-STRINGS ✨

Neat, formatted output

Code

```
name = "Sara"
price = 4.5

print(f"Hi {name}!")
print(f"RM {price:.2f}")
print(f"{name.upper()}")
```

Output

```
Hi Sara!
RM 4.50
SARA
```

:.2f means “2 decimal places”.

🧩 PUT IT TOGETHER

Palindrome checker

```
word = input("Enter a word: ")
clean = word.lower()

if clean == clean[::-1]:
    print("Palindrome! ✅")
else:
    print("Not a palindrome ❌")
```

A **palindrome** reads the same forwards and backwards.

level · racecar · madam

lower() makes “Level” work too.

🧠 QUICK CHECK

Predict before you run!

Q1. Which letter?

```
"Python"[1]
```

Q2. What piece?

```
s = "Hello World"
s[6:]
```

Q3. What is printed?

```
t = "hi"
t.upper()
print(t)
```

SUMMARY

What we learned today

Code (s = "Hello")	Result
s[0] · s[-1]	H · o
s[1:4]	ell
s[::-1]	olleH
s.upper() · s.lower()	HELLO · hello
s.replace("l", "L")	HeLLo
"ell" in s	True
f"{3.14159:.2f}"	3.14

💻 YOUR TURN

Practice time!

Open **exercises.ipynb** in the Day 6 folder and solve the 10 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ First, last and length of a word

⭐ ⭐ Email checker and username maker

⭐ ⭐ ⭐ Secret message cipher 🕵️

Next lesson: Lists 📋